

LAB EXPERIMENTS

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Experiment – 8 :

```
import pandas as pd
```

```
# Create DataFrame
```

```
sales_data = {
```

```
    "OrderDate":["1-6-18","1-23-18","2-9-18","2-26-18","3-15-18","4-1-18","4-18-18",
                "5-5-18","5-22-18","6-8-18","6-25-18","7-12-18","7-29-18",
                "8-15-18","9-1-18","9-18-18","10-5-18","10-22-18"],
```

```
    "Region":["East","Central","Central","Central","West","East","Central",
             "Central","West","East","Central","East","East","East",
             "Central","East","Central","East"],
```

```
    "Manager":["Martha","Hermann","Hermann","Timothy","Timothy","Martha","Martha",
              "Hermann","Douglas","Martha","Hermann","Martha","Douglas",
              "Martha","Douglas","Martha","Hermann","Martha"],
```

```
    "SalesMan":["Alexander","Shelli","Luis","David","Stephen","Alexander",
               "Steven","Luis","Michael","Alexander","Sigal","Diana",
               "Karen","Alexander","John","Alexander","Sigal","Alexander"],
```

```

"Item":["Television","Home Theater","Television","Cell Phone","Television",
        "Home Theater","Television","Television","Television",
        "Home Theater","Television","Home Theater","Home Theater",
        "Television","Desk","Video Games","Home Theater","Cell Phone"],

"Units":[95,50,36,27,56,60,75,90,32,60,90,29,81,35,2,16,28,64],

"Unit_price":[1198,500,1198,225,1198,500,1198,1198,1198,
              500,1198,500,500,1198,125,58.5,500,225]
}
df = pd.DataFrame(sales_data)

df["Sale_amt"] = df["Units"] * df["Unit_price"]

# Pivot Table
pivot = pd.pivot_table(df,
                        index="Item",
                        values="Units",
                        aggfunc="sum")

print("Item-wise Units Sold")
print(pivot)

```

```

# Pivot Table
pivot = pd.pivot_table(df,
                        index="Item",
                        values="Units",
                        aggfunc="sum")

print("Item-wise Units Sold")
print(pivot)

```

Item	Units
Cell Phone	91
Desk	2
Home Theater	308
Television	509
Video Games	16

Experiment – 9 :

```
import pandas as pd
```

```
# Create DataFrame
```

```
sales_data = {  
    "OrderDate":["1-6-18","1-23-18","2-9-18","2-26-18","3-15-18","4-1-18","4-18-18",  
                "5-5-18","5-22-18","6-8-18","6-25-18","7-12-18","7-29-18",  
                "8-15-18","9-1-18","9-18-18","10-5-18","10-22-18"],  
  
    "Region":["East","Central","Central","Central","West","East","Central",  
             "Central","West","East","Central","East","East","East",  
             "Central","East","Central","East"],  
  
    "Manager":["Martha","Hermann","Hermann","Timothy","Timothy","Martha","Martha",  
              "Hermann","Douglas","Martha","Hermann","Martha","Douglas",  
              "Martha","Douglas","Martha","Hermann","Martha"],  
  
    "SalesMan":["Alexander","Shelli","Luis","David","Stephen","Alexander",  
               "Steven","Luis","Michael","Alexander","Sigal","Diana",  
               "Karen","Alexander","John","Alexander","Sigal","Alexander"],  
  
    "Item":["Television","Home Theater","Television","Cell Phone","Television",  
           "Home Theater","Television","Television","Television",  
           "Home Theater","Television","Home Theater","Home Theater",  
           "Television","Desk","Video Games","Home Theater","Cell Phone"],  
  
    "Units":[95,50,36,27,56,60,75,90,32,60,90,29,81,35,2,16,28,64],  
  
    "Unit_price":[1198,500,1198,225,1198,500,1198,1198,1198,  
                 500,1198,500,500,1198,125,58.5,500,225]  
}
```

```
df = pd.DataFrame(sales_data)
```

```
df["Sale_amt"] = df["Units"] * df["Unit_price"]
```

```
# Pivot Table
```

```
pivot = pd.pivot_table(df,  
                        index=["Region","Manager","SalesMan"],  
                        values="Sale_amt",  
                        aggfunc="sum")
```

```
print("Region-wise, Manager-wise, Salesman-wise Total Sales")
```

```
print(pivot)
```

```
... Region-wise, Manager-wise, Salesman-wise Total Sales  
Sale_amt  
Region Manager SalesMan  
Central Douglas John      250.0  
          Hermann Luis    150948.0  
          Shellli      25000.0  
          Sigal      121820.0  
          Martha Steven    89850.0  
          Timothy David     6075.0  
East Douglas Karen      40500.0  
   Martha Alexander  231076.0  
          Diana      14500.0  
West Douglas Michael    38336.0  
   Timothy Stephen    67088.0
```

Experiment – 10 :

```
import pandas as pd
```

```
import numpy as np
```

```
# Generate random data
```

```
np.random.seed(10)
```

```
df = pd.DataFrame(
```

```
    np.random.randint(-50,51,size=(10,4)),
```

```
    columns=["A","B","C","D"]
```

```
)
```

```
print(df)
```

```
# Function for coloring
```

```
def color_numbers(val):
```

```
    if val < 0:
```

```
        return "color:red"
```

```
    else:
```

```
        return "color:black"
```

```
# Display styled dataframe
```

```
df.style.applymap(color_numbers)
```

```
...
   A  B  C  D
0 -41  50 -35  14
1 -22  39  43 -21
2 -42  23 -50 -10
3 -14 -34  50 -39
4  4  38  12 -17
5  22  50  28 -1
6  1  4  27  19
7 -37 -25 -37  42
8  36 -20 -20  39
9 -38  15 -19  7
/tmp/ipykernel_1207/2613384856.py:22: FutureWarning: Styler.applymap has been deprecated. Use Styler.map instead.
  df.style.applymap(color_numbers)

   A  B  C  D
0 -41  50 -35  14
1 -22  39  43 -21
2 -42  23 -50 -10
3 -14 -34  50 -39
4  4  38  12 -17
5  22  50  28 -1
6  1  4  27  19
7 -37 -25 -37  42
8  36 -20 -20  39
```